Frequency- and Spatial-Domain Saliency Network for Remote Sensing Cross-Modal Retrieval

Chengyu Zheng, Jie Nie, *Member, IEEE*, Bo Yin, *Member, IEEE*, Xiu Li, Yuntao Qian, *Senior Member, IEEE*, and Zhiqiang Wei, *Member, IEEE*

***Abstract*— Recently, remote sensing (RS) text–image retrieval has gained increasing attention among researchers due to its capability to provide abundant, inclusive, and multiperspective information. To extract the salient representation of the two modalities and realize information alignment, existing methods usually apply convolution calculation, matrix multiplication, and attention mechanisms to model features in the spatial domain. However, unlike natural images, RS data contain a considerable amount of noise, and the spatial-domain calculation methods will lead to information smoothing and noise exchange, ultimately reducing the robustness of feature representation. In this article, a frequency- and spatial-domain saliency network (FSSN) is proposed by extracting saliency features in both frequency and spatial domains to further enhance the efficiency of retrieval networks. Specifically, the FSSN first designs a frequency-domain intramodal low-pass filter (FILF) by using the Fourier transform (FT) to convert the spatial-domain representation into the frequency-domain representation, and leverage low-pass filtering to filter out the noise information of the image. Afterward, the space-domain intermodal boundary-based saliency (SIBS) module is devised, which fully utilizes the positive and negative boundaries and draws an innovative assessment mechanism to automatically recognize the effective regions and words without noise shifting in the spatial domain. Finally, the frequency- and spatial-domain saliency fusion (FSSF) is designed to realize the effective integration of features obtained from the frequency and spatial domain according to the consistency of the identity masks. Quantitative and qualitative experiments are implemented on four RS benchmarks to showcase the notable effectiveness achieved by the joint modeling network of multiple domains. Specifically, the proposed FSSN outperforms the best model RemoteClip by 7.58% of the mR evaluation matrix on the UCM dataset.**

***Index Terms*— Cross-modal retrieval, remote sensing (RS), spatial and frequency.**

I. Introduction

REMOTE sensing (RS), as an emerging technology, has significantly improved human comprehension of geographical space [1], [2], [3]. However, RS data have complex characteristics, such as large scale, intangible noise, and low resolution, thus, obtaining valuable information from enormous RS data has gradually become a popular topic. As one of the important manners of data processing, cross-modal retrieval [4], [5], [6] could fully integrate multiple modal data, and utilize mutual guidance and complementation to enhance the reliability and availability of RS information. Wherein, RS images and text retrieval (RSITR) is the core task of cross-modal retrieval since it can consider both the comprehensiveness of images and the intuitiveness of texts, and explore the potential correlation between multiple models to further improve the utilization of RS data, which has aroused increasing interest among more and more researchers.

Recently, beneficial to the continuous sprouting of convolutional neural networks (CNNs), able to mine the potential semantics of images and texts, the RS cross-modal retrieval models have made significant advances. The CNN-based image–text retrieval framework generally utilizes CNNs, like ResNet [7] for image and GRU [8] for text, to extract high-level features first, which are then subjected to similarity measurement to obtain the best matching pair. As expected, it is more challenging to perform cross-modal retrieval tasks on RS data than natural data since the former contains more complex characteristics, like the large-scale span, intangible noise, and scattered targets in RS images. Therefore, considerable work has been dedicated to improving the accuracy of RS cross-modal retrieval. For example, taking the multiscale property of RS objects into account, the work [9] adapted multiscale feature input and dynamic noise filtering technology. In recent years, the rapid development of pretrained large models, such as CLIP [10], MiniGPT-4 [11], RemoteCLIP [12], and GeoRSCLIP [13], has provided new ideas for RS cross-modal retrieval. Building upon CLIP, PETL [14] proposed a multimodal RS adapter to effectively and efficiently transfer visual-language knowledge from the natural domain to the RS domain.

Most RSITR networks are dedicated to exploring salient information in RS data, defined as two types: intramodal saliency and intermodal saliency. For intramodal saliency, current methods are primarily categorized into three aspects: semantic-level, target-level, and knowledge guidance. The semantic-level methods usually involve disentanglement [15], [16], CNNs [17], [18], or self-attention mechanisms [19], [20] to highlight effective semantic information from complex RS environments. However, the accuracy of these methods is compromised when it comes to the edges or details due

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Chengyu Zheng, Jie Nie, Bo Yin, Xiu Li, and Zhiqiang Wei are with the Faculty of Information Science and Engineering, Ocean University of China, Qingdao 266100, China (e-mail: niejie@ouc.edu.cn).

Yuntao Qian is with the College of Computer Science, Zhejiang University, Hangzhou 310027, China.

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to the low resolution of RS images and the high similarity of targets. As a result, some studies focus on target-level saliency. For instance, in the study [21], PP-YOLO [22] was introduced into the RS image–text retrieval task to differentiate targets in RS images and achieve more detailed modeling. Besides, knowledge guidance [23], [24] is another way to mine intramodal saliency, as it allows the removal of noise present within the RS target under the supervision of robust knowledge. Despite the significant contribution of intramodal saliency methods, they often suffered limited performance because of the inherent expression gaps in different modalities and the misalignment of data. Consequently, intermodal saliency is gradually becoming mainstream. One effective approach is to utilize cross-attention in the Transformer decoder structure, which could identify potential correlations among cross-modal data by calculating similarities, to achieve cross-modal saliency modeling [25], [26], [27]. In addition, some researchers [28], [29] created various intermediate representations to extract cross-modal saliency.

Despite the prominent value and profound influence achieved by these saliency methods, they are summarized as "spatial-domain" methods since they only implement convolution calculations, matrix multiplication, or dimension changes to extract high-level features in the spatial domain. However, unlike natural images, RS data are captured by receivers far away from ground objects, thus containing a lot of noise, such as ineffective targets and atmospheric scattering. If the retrieval model still applies spatial-domain calculation methods, these noises will be transferred during the feature extraction process, reducing the robustness of feature representation. Also, these spatial-domain calculation methods often involve information exchange, which can result in information smoothing or overshadowing of small objects during the training process. For example, in Fig. 1, the small object "swimming pool" is overwhelmed by large objects "tennis court" and "road," ultimately leading to inaccurate results.

To tackle these downsides, we propose a frequency- and spatial-domain saliency network (FSSN) for the RSITR task to further enhance the efficiency of the retrieval model. We still adopt the traditional RSITR framework to extract high-level feature representation for images and texts, respectively. In particular, we innovatively devise three modules, named frequency-domain intramodal low-pass filter (FILF), space-domain intermodal boundary-based saliency (SIBS), and frequency- and spatial-domain saliency fusion (FSSF), to achieve robust feature extraction and matching, so as to ensure effective and reliable image–text retrieval in dealing with RS data that contains lots of noises. Specifically, the FILF is designed to fully utilize low-frequency signals to filter noise in images by exploiting multiple frequency denoising layers containing two branches, one of which uses a low-pass filter to filter the ineffective noise, and the other is complementary to ensure information retention by applying CNN. Besides, the SIBS is proposed to continue mining effective information in the spatial domain by involving similarity distribution boundaries of matched and mismatched sample pairs. Based on these boundaries, we devise novel image and text assessment mechanisms (TAMs) that can automatically

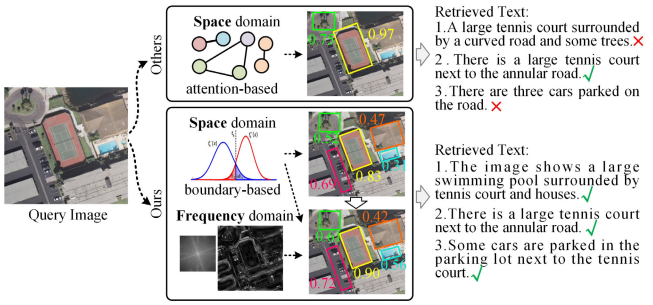


Fig. 1. Illustrates the comparison of existing methods and ours. Existing attention-based spatial-domain strategies often involve information exchange, which can result in information smoothing or noise transferring during the training process. In contrast, our approach combines boundary-based spatial domain and low-pass filter-based frequency-domain techniques to explore salient regions. This enables the model to extract information from multiple perspectives and remain unaffected by noise, thus yielding accurate ranking results.

generate spatial confidence masks, implying the effectiveness of the features. These confidence masks are then converted into spatial identity masks, recording positive regions or words, which are integrated with the original feature to filter the useless noise. Finally, the FSSF module fuses the image frequency- and spatial-domain salient features according to the consistency of the identity mask. As shown in Fig. 1, compared with the existing attention-based methods for saliency mining, our approach utilizes a boundary modeling mechanism to recognize the effective areas of RS data without information transfer, so as to minimize disturbance from noise and obtain more robust salient region outputs. In addition, the joint modeling of space and frequency domains could realize complementation, allowing for more accurate information mining and correct ranking results. The effectiveness of the proposed architecture is demonstrated through experiments conducted on four RSITR datasets. The following points outline the key contributions of this article.

1) We propose an RSITR network FSSN that considers both frequency-domain and spatial-domain saliency modeling, so as to ensure effectiveness and reliability in dealing with complex RS data.
2) We propose the FILF, which could fully utilize low-frequency signals to filter noise in images. Besides, we devise the SIBS module by applying the optimal boundary of image and text to distinguish and mine the effective salient information in images and texts without noise information transfer. Finally, the FSSF is designed to integrate the spatial- and frequency-domain features effectively.
3) Quantitative and qualitative experiments are implemented to confirm the effectiveness of our architecture on four RS text–image retrieval datasets.

The rest of the article is organized as follows. Section II presents the related works on cross-modal retrieval of RS image–text. Section III provides the details of the proposed FSSN. The corresponding experimental results and analysis are described in Section IV. Finally, we discuss the contributions of this article and conclude in Section V.

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II. Related Work

A. RS Images and Text Retrieval

The RSITR is a process where a retrieval network is required to match a given query image (or text) with the corresponding labeled text (or image) in a database based on similarity. The earliest derived cross-modal retrieval methods were automatic image captions [30], [31], [32], which could generate an accurate sentence description from a given image, beneficial to humans to quickly interpret images, and gradually developed into a research hotspot. Lately, there has been an increasing interest in research related to RSITR. Abdullah et al. [33] introduced a bidirectional network by building a new dataset and a novel paradigm to achieve the RSITR. Different from cameras-captured natural images, RS images possess several unique characteristics, like large-scale span, scattered targets, complex spatial structure, and low resolution, thus bringing great challenges to RSITR. For instance, to address the multiscale problem, Yuan et al. [9] devised the AMFMN network, which simultaneously extracts both small-scale and large-scale features. Considering the redundant targets in the RS images, Yuan et al. [34] introduced a lightweight network to reduce occupancy and overhead. In addition, some methods are also committed to optimizing the loss function. For example, Li et al. [35] devised a false negative sample elimination mechanism specific for the triple loss to more effectively reduce the distance of positive samples while increasing the distance of negative samples, benefiting the model training process. Moreover, to better extract foreground targets in RS images with complex background noise, Zhu et al. [36] established a multitask joint learning framework, which incorporates segmentation into the cross-modal retrieval task, outputting more ideal results.

Recently, significant advancements have been made in pretrained large models, such as CLIP [10], BLIP [37], and MiniGPT-4 [11], which have become a hot research topic. These models demonstrate strong generalization and feature representation capabilities by being trained on a large amount of annotated data across multiple tasks. In addition, considering the uniqueness of RS data, specific pretrained large models, like RemoteCLIP [12] and GeoRSCLIP [13], have been developed. As a result, there is a growing interest in applying pretrained large models to image–text retrieval tasks. For example, Liu et al. [38] utilized CLIP to encode RS data, and applied text as a medium to bridge the natural domain with the RS domain. Yuan et al. [14] proposed a multimodal RS adapter to effectively and efficiently transfer visual-language knowledge from the natural domain to the RS domain.

B. Saliency Modeling in the Spatial Domain

The saliency modeling in the spatial domain is mainly divided into two types, termed intramodal saliency and intermodal saliency. One method for capturing intramodal saliency is to achieve semantic emphasis through disentanglement [15], [16], CNNs [17], [18], or self-attention mechanism [19], [20]. For instance, Pan et al. [18] applied convolution kernels of various sizes to identify prominent objects at a detailed level. In addition, Cheng et al. [19] designed a multiattention mechanism to improve the discriminative capability of visual features in RS images within complex scenes. Considering the Transformer can capture long-distance dependencies from a global view, Zhang et al. [20] integrated it with CNN to extract spatial interregion relations of images and long-term interword relations of texts. However, the semantic emphasis method yields unsatisfactory results on the edges or details due to the low resolution of RS images and the extremely high similarity of targets. Therefore, Yuan et al. [21] first introduced PP-YOLO [22] into the RS image–text retrieval task to highlight targets in RS images and realize fine-granularity modeling. Yao et al. [39] further explored the relationships among various objects by utilizing hypergraph learning. Besides, knowledge guidance [23], [24] is another way to mine intramodal saliency, as it allows the removal of noise present within the RS target under the supervision of this knowledge. For example, Ji et al. [23] established a text knowledge base by the process of knowledge initialization, construction, filtration, and alignment to enable the model to concentrate on key concepts more effectively.

Despite the prominent contribution of intramodal saliency methods, they often suffered limited performance because of the inherent expression gaps in different modalities and the misalignment of data. Consequently, intermodal saliency is gradually becoming mainstream. One effective approach is to utilize cross-attention in the Transformer decoder structure, which could identify potential correlations among cross-modal data by calculating similarities, to achieve cross-modal saliency modeling [25], [26], [27]. As a typical representative, Lee et al. [25] introduced the stacked cross-attention network (SCAN), which applied cross-attention to emphasize important regions or words and then compared them with the original regions or words. Besides, Tang et al. [27] fully exploited the advantages of the Transformer and adapted it to the field of RS, which models the potential relationships between different modalities. In addition, some researchers created various intermediate representations to extract cross-modal saliency. For instance, Cheng et al. [28] and Lv et al. [29] combined features from different modalities as intermediate representations to achieve data alignment. Meanwhile, based on the image and text branches, Ma et al. [40] introduced the RoT architecture as an additional branch, to explore effective multiple-modal information. To further learn subtle image–text relationships, Zhao et al. [41] designed a masking-based interaction framework by hiding certain information of the data.

Although the above saliency modeling methods in the spatial domain have greatly promoted the improvement of RS cross-modal retrieval accuracy, there are still unsatisfactory results due to these networks only implementing convolution calculations, matrix multiplication, or dimension changes to extract high-level features in the spatial domain. However, RS data are captured by receivers far away from ground objects, thus containing more inevitable noise, such as ineffective targets and atmospheric scattering. If the retrieval model still applies spatial-domain calculation methods, these noises will be transferred during the feature extraction process, reducing the robustness of feature representation. Therefore,

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this article aims to address this issue and propose a more effective and reliable image–text retrieval network, namely, FSSN, by introducing both frequency- and spatial-domain saliency extraction methods to generate a more robust description of multimodal data. In fact, our method made some important innovations. In the FILF module, we advocate for fully utilizing low-pass filtering to remove noise from images. Currently, in RS cross-modal retrieval tasks, there is no relevant method that adopts frequency-domain signals for feature extraction. Besides, the frequency-domain extraction method is unique in that it includes two branches, one of which involves an affine point-wise map of a CNN in the physical space to ensure the information transfer, and the other one applies the FT to convert the feature into the frequency domain and uses a low-pass filter to filter the ineffective noise. In the SIBS module, based on the similarity distribution boundary, we innovatively create the image and TAM to automatically generate identity masks, implying the effective areas and words of the features. We believe that in the RS cross-modal retrieval task, we are the first to consider spatial-domain and frequency-domain features and achieve their effective fusion.

III. Proposed Method

Here, we introduce the FSSN designed to enhance the accuracy of RSITR. We begin with an overview of the FSSN, highlighting its general motivation and architecture. Next, we will explore the complex components of the proposed modules in great detail. Finally, we illustrate a detailed explanation of the loss function used to train the network.

A. Overview of the Proposed FSSN

In order to reduce noise interference and obtain robust saliency features from complicated RS data to achieve accurate retrieval matching, we propose a novel network, namely, FSSN, which innovatively designs FILF, SIBS, and FSSF modules. The FILF fully utilizes low-frequency signals to filter noise in images by exploiting multiple frequency denoising layers containing a low-pass filtering branch and a CNN branch. The SIBS applies the similarity distribution boundaries of matched and mismatched sample pairs, and devises novel image and TAMs to distinguish the effective salient information in images and texts without noise information transfer. Finally, FSSF is designed to effectively integrate the spatial- and frequency-domain features according to the consistency of the identity mask.

Specifically, the architecture of FSSN is shown in Fig. 2. As mentioned above, the RSITR is that, given a query image (or text), the retrieval network needs to perform the matching operation with the text (or image) in the database to retrieve the best matching pairs. Here, we formulate the above process for better understanding. Let $\mathcal{D} = \{I, T\}$ be defined as the images and texts from the database. We first apply pretrained feature extractors [42], [43] to transform I and T into a high-level feature representation $V \in \mathbb{R}^{N\times D}$ and $U \in \mathbb{R}^{L\times D}$, respectively. Subsequently, the generated features are fed into two different modules, namely, FILF and SIBS, designed for modeling in the frequency and spatial domain, respectively. The FILF module employs multiple frequency denoising layers $\mathcal{L}(\cdot)$, where the Fourier transform (FT) is exploited to convert image features from the spatial domain into the frequency-domain to filter the noise in V, and then inverse FT (IFT) is applied to recover spatial-domain representation, generating an image frequency saliency feature V^f. The SIBS module devises the image assessment mechanism (IAM) and TAM through the participation of similarity distribution boundaries t_k^v and t_k^u of image and text to generate the image and text spatial identity masks $\widehat{m}^{vs}$ and $\widehat{m}^{us}$, these masks are then utilized to obtain spatial saliency features V^s and U^s of image and text. Afterward, FSSF first utilizes the calculating identity mask (CIM) to obtain image frequency identity mask $\widehat{m}^{vf}$. According to the generated $\widehat{m}^{vf}$ and $\widehat{m}^{vs}$, the image frequency and spatial saliency feature V^f and V^s are fused by performing the IAM again to calculate the final image saliency features V^o. Finally, the V^o and U^s are arranged to compute the similarity scores, which are integrated into the loss function to participate in the training of the model.

B. Feature Extraction

We will detail the feature representation in two aspects, including image feature extraction and text feature extraction.

1) Image Feature Extraction: In recent years, CLIP [10] has exhibited strong feature representation capabilities as a result of its training on extensive datasets, utilizing over 400 million parameters. Furthermore, in response to the unique characteristics of RS data, RemoteCLIP [12] was developed. It is built upon CLIP and trained on seventeen diverse and extensive RS datasets across three tasks. Its framework integrates vision Transformer (VIT) [42] for images and Transformer [43] for text, proving to be effective in various downstream RS tasks.

Thus, we utilize pretrained VIT on RemoteCLIP as our image feature extractor. Specifically, RS image $I \in \mathbb{R}^{H\times W\times 3}$ is first divided into fixed-sized patches. Then, these patches are encoded by VIT to get an image feature $V \in \mathbb{R}^{N\times D}$

$$V = \mathcal{F}\left(I, \theta^I\right) \tag{1}$$

where $\mathcal{F}(\cdot)$ denotes the vision encoder, θ^I is the frozen weights.

2) Text Feature Extraction: As previously mentioned, the Transformer trained in RemoteCLIP is very beneficial for text extraction, so we apply it as our text extractor. It is important to note that masked self-attention was utilized in the text encoder to maintain the ability to incorporate language modeling as an auxiliary objective. More specifically, a sentence T is first embedded as word vectors $e(T)$, which is delivered to the Transformer to obtain text feature $U \in \mathbb{R}^{L\times D}$

$$U = \mathcal{T}\left(T, \theta^T\right) \tag{2}$$

where $\mathcal{T}(\cdot)$ denotes the text encoder, θ^T is the frozen weights. Note that Trans in Fig. 2 is the abbreviation of Transformer.

C. FILF Module

As mentioned above, feature extraction only in the spatial domain is a suboptimal choice for retrieval models, especially when dealing with RS data containing a large amount of noise.

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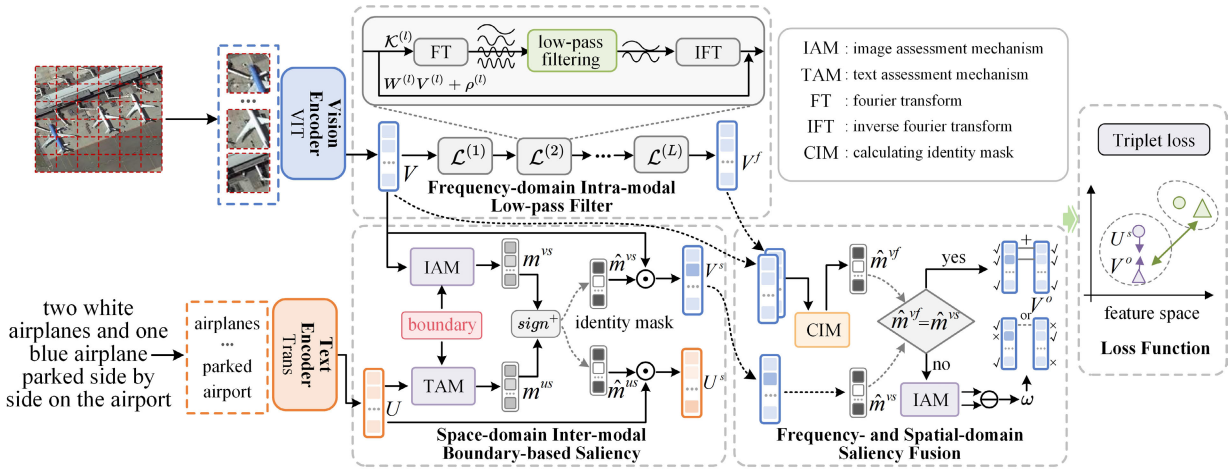


Fig. 2. Architecture of the proposed FSSN. First, we apply pretrained feature extractors to explore the feature representation of the image and text, respectively. The generated features are fed into different modules, namely, FILF and SIBS. The FILF fully utilizes low-frequency signals to filter noise in images by exploiting multiple frequency denoising layers. The SIBS applies the similarity distribution boundaries of matched and mismatched sample pairs, and devises novel image and TAMs to distinguish the effective salient information in images and texts without noise information transfer. Finally, FSSF is designed to effectively integrate the spatial- and frequency-domain features according to the consistency of the identity mask.

To solve this problem, it is considered that the low-frequency signals in the frequency-domain represent the primary features, such as the "tennis court" and "cars" in the RS image of Fig. 1, while the high-frequency signals in the frequency domain refer to the regions of dramatic changes in the RS image, in other words, is the noise and other useless information, thus we believe that low-pass filtering should be fully utilized to filter noise in images. Therefore, based on the above analysis, the FILF module is designed by using the FT to convert the spatial-domain representation into the frequency-domain representation, and leveraging low-pass filtering to filter out the noise information of the image, which is then fed into IFT to restore the original feature expression. The specific process is as follows.

Motivated by the work [44], achieving an accelerated solution of partial differential equations (PDEs) by exploring the Fourier representation of flow fields and establishing a bridge between pure machine learning approaches and the best numerical or hybrid solvers, we decide to perform several frequency denoising layers $\mathcal{L}^{(l)}$ on the original image feature V

$$V^f = \mathcal{G}\left(\mathcal{L}^{(L)} \circ \cdots \circ \mathcal{L}^{(1)}(V)\right) \tag{3}$$

where $\circ$ indicates function composition, L is the number of layers/iterations, $\mathcal{L}^{(l)}$ means the l'th frequency denoising layers, and $\mathcal{G}$ denotes the projection operator that maps the last latent representation $V^{(L)}$ to the output. $\{V^f\}_{n=1}^{N}$ is the image frequency saliency feature, which is also the output of the FILF module. Each layer $\mathcal{L}^{(l)}$ consists of two terms

$$\mathcal{L}^{(l)}\left(V^{(l)}\right) = \sigma\left(W^{(l)} \odot V^{(l)} + \rho^{(l)} + \mathcal{K}^{(l)}\left(V^{(l)}\right)\right) \tag{4}$$

where σ is an activation function. Equation (4) mainly consists of two branches, one of which is the item $W^{(l)}V^{(l)} + \rho^{(l)}$, representing an affine point-wise map of a CNN in the physical space to ensure the information transfer, and $W^{(l)}$ is the training weight and $\rho^{(l)}$ is the bias. The other $\mathcal{K}^{(l)}$ is a kernel integral operator using the FT, which first maps feature V to the frequency-domain space, where applying low-pass filters and convolution operations to filter out the high-frequency signals, and then the generated features are mapped back to the spatial domain

$$\begin{aligned} V_f^{(l)} &= \mathrm{FT}\left(V^{(l)}\right) \\ V_c^{(l)} &= W_f^{(l)}\left(\text{Low Filter}\left(V_f^{(l)}\right)\right) + \rho_f^{(l)} \\ \mathcal{K}^{(l)} &= \mathrm{IFT}\left(V_c^{(l)}\right) \end{aligned} \tag{5}$$

where FT represents the Fourier transform, and IFT means the inverse Fourier transform. Low Filter refers to a low-pass filtering operation, which can filter the noise in the image and retain effective information. $W_f^{(l)}$ is the training weight of CNN and $\rho_f^{(l)}$ stands for the bias. Through the above process, we finally obtain a robust image frequency feature V^f.

D. SIBS Module

In order to achieve cross-modal feature alignment and noise filtering, most existing methods are willing to adopt an attention mechanism since it can activate the saliency of the current modality according to the saliency of other modalities by information guidance and transfer. However, as we mentioned, RS images contain a considerable amount of noise, and the information transfer will lead to the noise being exchanged between multiple modal data, which not only fails to reduce the noise, but even increases the noise, and that result is incompatible with our original purpose. After considering the problem, we thought of how to mine the effective salient information in images and texts without information transfer. Fortunately, previous work NAAF [45] devised sampling strategies and probability distribution difference maximization mechanisms to learn the similarity distribution boundaries of text to distinguish the positive and negative representation. Therefore, referring to NAAF, we first introduce boundaries to our module and design the image and TAM to recognize the effective or positive regions and words, which are

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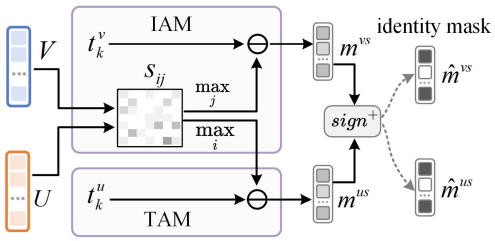


Fig. 3. Illustration of spatial identity mask. Based on the similarity distribution boundaries t_k^v and t_k^u, the image and text features are sent into IAM and TAM to generate spatial confidence masks. Then, these spatial confidence masks are converted into identity masks $\widehat{m}^{vs}$ and $\widehat{m}^{us}$ by sign$^+$, recording positive regions or words of images and texts.

then recorded in the spatial identity masks. These masks are exploited to be integrated with the original data to obtain final spatial salient features.

1) Spatial Identity Mask: The generation process of the spatial identity mask is displayed in Fig. 3. The first step in obtaining the spatial identity mask is to compute the boundaries, here, we follow the research NAAF and illustrate the image boundary as an example. Regarding the mismatched and matched region-word fragment pairs, the sample operation is first conducted to group the similarity degree sets as S_k^+ and S_k^-, where S_k^+ and S_k^- are the sets of matched region-word similarity and mismatched region-word similarity, respectively. According to the sampled two sets, the mismatched and matched probability distributions $f_k^+(s)$ and $f_k^-(s)$ of region-word fragment pairs can be obtained. Subsequently, based on the probability distributions, an optimal image boundary t_k^v is learned to maximally distinguish the two distributions. Note that t_k^v is continuously updated with the training index k, and then used as a kind of indicator for the image, interpreting as when the similarity between the image region and a certain word in the sentence is higher than the boundary, this region is considered to be a saliency feature, and vice versa (the generation process of text boundary t_k^u is similar to t_k^v, and we don't elaborate again for simplicity. A more detailed explanation of the boundary calculation process can be found in NAAF [45]).

In order to utilize t_k^v to extract saliency features, we devise an IAM, denoted as $\mathcal{P}^v$, which holds the capability to automatically generate the image spatial confidence mask m^{vs} by applying the V_i and t_k^v. The m^{vs} determines the effectiveness of the feature, the positive values in m^{vs} imply saliency features, and negative numbers are noise. The $\mathcal{P}^v$ also requires the assistance of the text feature U_j

$$\begin{aligned} m_i^{vs} &= \mathcal{P}^v\left(t_k^v, V_i, U_j\right) \\ \mathcal{P}^v\left(t_k^v, V_i, U_j\right) &= s_i - t_k^v \end{aligned} \tag{6}$$

where s_i represents the maximum similarity between a region i and all words, $s_i = \max_j(\{s_{ij}\}_{j=1}^l)$, where s_{ij} is computed as

$$s_{ij} = \frac{V_i \times U_j^T}{\|V_i\|\|U_j\|} \tag{7}$$

where s_{ij} is the similarity between the image V_i and the text U_j. $\times$ refers to matrix product operation. V_i is the region feature of image V and U_j is the word feature of text U. Finally, m^{vs} is converted into an image spatial identity mask $\widehat{m}^{vs}$ by sign$^+$, which could record positive numbers as 1 and negative numbers as 0

$$\widehat{m}^{vs} = \text{sign}^+(m^{vs}). \tag{8}$$

Similarly, the TAM is designed with the same strategy, by feeding text feature U, text boundary t_k^u, and image feature V into $\mathcal{P}^u$, the text spatial confidence mask m^{us} is generated automatically, then converted into text spatial identity mask $\widehat{m}^{us}$ by sign$^+$.

2) Spatial Saliency Extraction: Following $\widehat{m}^{vs}$ and $\widehat{m}^{us}$, we can recognize the effective information and noise in the image and text. Then, we integrate these identity masks with the original features V and U to filter the useless noise

$$\begin{aligned} V^s &= \widehat{m}^{vs} \odot V \\ U^s &= \widehat{m}^{us} \odot U. \end{aligned} \tag{9}$$

The $V^s \in \mathbb{R}^{N\times D}$ and $U^s \in \mathbb{R}^{L\times D}$ are the image and text spatial saliency features.

E. FSSF Module

There is ambiguity in the salient regions in images calculated by the spatial and frequency domain, thus, the FSSF module is devised to achieve accurate and effective feature fusion. However, in the frequency domain, there is a lack of indication to tell which areas are filtered, so as to achieve effective fusion with the features of the spatial domain. Considering that the identity mask in the spatial domain can clearly indicate the valid areas, we also calculate an identity mask for the frequency domain. According to the consistency of these two masks, we can effectively integrate the image features from two domains.

1) Calculating Frequency Identity Mask: Since we cannot go inside the low-pass filter to check which regions have been filtered, we think of comparing the generated image frequency feature V^f with the original feature V. The regions with large discrepancies are considered the filtered part, which is defined as noise, thus $\widehat{m}^{vf}$ is assigned a value of 0, otherwise, the regions have not been filtered out by a low-pass filter, and must contain lots of effective information, so $\widehat{m}^{vf}$ is assigned a value of 1. First, we calculate the discrepancies γ_i between V^f and V

$$\gamma_i = \left|V_i^f - V_i\right| \tag{10}$$

where $|\cdot|$ denotes the absolute value operation. Then, the identity mask $\widehat{m}^{vf}$ is formulated as follows:

$$\begin{cases} \widehat{m}_i^{vf} = 1, & \text{if } |\gamma_i| < \varepsilon \\ \widehat{m}_i^{vf} = 0, & \text{else} \end{cases} \tag{11}$$

where ε is a hyperparameter whose value is set in subsequent experiments.

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2) Image Saliency Fusion: After obtaining the saliency masks of the two domains, the fusion operation can be performed. When the frequency identity mask $\widehat{m}^{vf}$ of the image is consistent with the spatial identity mask $\widehat{m}^{vs}$, we perform an element addition operation to fuse the spatial and the frequency features. Otherwise, the IAM $\mathcal{P}^v$ is carried out again to decide features with high confidence, which are utilized to calculate the final image features. Specifically, the image feature V^o is computed through element addition between V^f and V^s if identity masks $\widehat{m}_i^{vf}$ and $\widehat{m}_i^{vs}$ are consistent

$$V^o = V^f + V^s. \tag{12}$$

In another case, we first use the IAM $\mathcal{P}^v$ to evaluate the confidence of the two features with the assistance of the text spatial feature U^s

$$\begin{aligned} m_i^{of} &= \mathcal{P}^v\left(t_k^v, V_i^f, U_j^s\right) \\ m_i^{os} &= \mathcal{P}^v\left(t_k^v, V_i^s, U_j^s\right) \end{aligned} \tag{13}$$

where m_i^{os} and m_i^{of} refer to image spatial and frequency confidence masks, respectively. Then, based on the confidence, a comparison is considered to select features with high confidence as the final image feature, which is formulated as

$$\omega_i = \left|m_i^{of}\right| - \left|m_i^{os}\right| \tag{14}$$

where ω_i is the difference between the two confidence masks. The positive value of ω indicates that the spatial features are more reliable, whereas the frequency-domain features are considered more trustworthy

$$\begin{cases} V^o = \widehat{m}^{vf}(V^f + \sigma V^s), & \text{if } \omega > 0 \\ V^o = \widehat{m}^{vs}(\sigma V^f + V^s), & \text{else} \end{cases} \tag{15}$$

where $V^o \in \mathbb{R}^{N\times D}$ is the fused image saliency feature, and σ represents the smoothing factor, which takes a very small value. We can deduct from the above equation that if the confidence of V^f is greater than V^s, $\|m_i^{of}\| > \|m_i^{os}\|$, resulting in $\omega_i > 0$, then V^o gets the value from V^f, and vice versa.

F. Loss Function

As usual, the triplet loss is applied to train our cross-modal retrieval model since it can achieve a more refined measure of difference. The triple loss aims to minimize the distance between features of the same label and maximize the distance between features of different labels. We utilize bidirectional triple loss proposed by [46]

$$\begin{aligned} \mathcal{L}(I,T) = &\sum_{\hat{U}^s}[\alpha - S(V^o, U^s) + S(V^o, \hat{U}^s)]_+ \\ &+ \sum_{\hat{V}^o}[\alpha - S(V^o, U^s) + S(\hat{V}^o, U^s)]_+ \end{aligned} \tag{16}$$

where α is the margin and $[\cdot]_+ = \max(\cdot, 0)$. $S(V^o, U^s)$ represents the similarity between image and text with the same labels. $S(V^o, \hat{U}^s)$ and $S(\hat{V}^o, U^s)$ represent the similarity of images and sentences with different labels.

IV. Experimental Results and Analysis

We will be performing experiments to assess the effectiveness of FSSN. Meanwhile, we introduce the datasets, baseline, and implementation details used in the experiment.

A. Datasets, Baselines, and Evaluation Metrics

We conduct experiments on four RSITR task benchmarks: UCM, Sydney, RSICD, and RSITMD. The UCM dataset, built by [47], comprises 21 categories of land-use images, each containing 100 images. The Sydney dataset, based on [47], contains 613 images. The RSICD dataset [31] was used for the captioning task and included a total of 10 921 RS images. The RSITMD dataset, provided by [9], derives from the RSICD dataset and includes 23 715 captions for 4743 images. All the images are described in five texts.

As for baselines, 18 image–text retrieval methods are compared, including five classic image–text retrieval networks applied on natural data, abbreviated as "N," which are SCAN [25], CAMP [48], MTFN [49], SGRAF [50], NAAF [45], and 13 networks dedicated to RS image–text retrieval, termed "R," including SAM [28], LW-MCR [34], AMFMN [9], GaLR [21], MCRN [51], CMFM [52], PIR [24], HVSA [53], KAMCL [23], DOVE [40], IEFT [27], RemoteCLIP [12], and GLISA [54]. It should be noted that since some methods do not have publicly available codes, we directly quote the results provided by the authors in the published manuscript and mark them with the symbol "*."

We perform two types of RSITR tasks: 1) retrieving relevant sentences for a given query image (image-to-text) and 2) retrieving relevant images for a given query text (text-to-image). We utilize the metric of recall at K ($R@K$) with K set to 1, 5, and 10 to assess the designed modules. This metric signifies the proportion of ground truth that appears in the top K results. In addition, we calculate the mean recall rate of R@K, denoted as mR, following the suggestion of Huang et al. [55]. This metric offers a more comprehensive and insightful evaluation of the model's performance.

B. Implementation Details

In this section, we will explain the details involved in implementing our network. During preprocessing, RS images are smartly resized to 224 × 224 (keeping aspect ratio), augmented with random rotations/flips/color adjustments, and normalized using ImageNet standards. Text data is cleaned, tokenized into subwords, and adjusted to fixed lengths. The D is 512 in feature V and U. For the triple loss (16), we take the value of margin threshold α to 0.2. σ is set to 0.1 in the proposed network FSSN. ε is set to 0.5. All models have been conducted using PyTorch and the Adam optimizer with a learning rate of 0.0002. The model has been trained for 30 epochs with a batch size of 25. The experiments have been conducted on a server equipped with one Tesla V100 SVM2 16-GB GPU.

C. Comparison With State-of-the-Art Methods

In this section, we present a comparison of our method with 18 baseline models across four different datasets. The results

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TABLE I
COMPARISONS OF IMAGE–TEXT RETRIEVAL EXPERIMENTS ON RSICD AND RSITMD DATASETS

Type	Method	Backbone Vision/Text	RSICD			RSITMD		
			Image-to-Text R@1 / R@5 / R@10	Text-to-Image R@1 / R@5 / R@10	mR	Image-to-Text R@1 / R@5 / R@10	Text-to-Image R@1 / R@5 / R@10	mR
N	SCAN t2i	ResNet/GRU	4.42 / 11.20 / 17.68	4.02 / 11.54 / 18.60	11.24	10.59 / 28.72 / 38.41	10.04 / 29.54 / 42.91	26.70
	SCAN i2t	ResNet/GRU	5.90 / 13.21 / 19.96	3.86 / 16.83 / 26.49	14.38	10.84 / 25.86 / 37.26	9.62 / 29.80 / 41.06	25.74
	CAMP-triplet	ResNet/GRU	5.22 / 13.05 / 21.02	4.30 / 17.10 / 27.57	14.71	11.82 / 27.40 / 38.12	8.69 / 27.17 / 43.60	26.13
	CAMP-bce	ResNet/GRU	4.50 / 10.08 / 16.48	3.03 / 15.12 / 23.05	12.04	9.21 / 22.56 / 35.73	6.81 / 25.65 / 40.05	23.34
	MTFN	ResNet/GRU	5.01 / 12.86 / 21.30	4.96 / 12.14 / 29.12	14.23	10.80 / 27.68 / 36.40	9.82 / 30.28 / 48.27	27.21
	SGRAF	ResNet/GRU	3.93 / 13.36 / 22.78	3.42 / 14.53 / 26.72	14.12	5.97 / 18.81 / 30.09	6.02 / 24.25 / 41.81	21.16
	NAAF	ResNet/GRU	5.92 / 16.31 / 27.23	4.21 / 17.20 / 26.33	16.20	10.50 / 30.75 / 41.86	9.13 / 30.38 / 44.25	27.81
R	SAM i2t	CNN/RNN	6.41 / 20.10 / 32.89	5.91 / 18.08 / 29.73	18.85	13.27 / 30.09 / 44.91	9.87 / 33.89 / 50.66	30.45
	SAM t2i	CNN/RNN	5.91 / 18.59 / 30.63	4.82 / 18.10 / 33.44	18.58	11.73 / 27.25 / 40.06	11.07 / 35.03 / 50.94	29.35
	LW-MCR	SqueezeNet/RNN	3.66 / 13.63 / 21.68	3.77 / 15.32 / 28.20	14.38	6.64 / 17.92 / 30.31	5.53 / 25.51 / 44.34	21.71
	AMFMN	ResNet/GRU	5.39 / 15.32 / 28.82	5.05 / 18.19 / 29.07	16.97	11.37 / 27.70 / 39.29	9.04 / 32.46 / 49.68	28.26
	GaLR	ResNet+ppyolo/GRU	4.67 / 17.47 / 27.63	4.65 / 20.20 / 34.80	18.24	13.27 / 31.85 / 41.15	10.92 / 35.3 / 52.96	30.91
	MCRN	CNN/GRU	5.58 / 18.38 / 30.56	4.87 / 18.92 / 33.25	18.59	8.85 / 23.67 / 35.84	8.05 / 30.97 / 49.60	26.16
	CMFM	ResNet/GRU	5.40 / 16.47 / 27.54	4.15 / 17.99 / 31.25	17.13	8.85 / 23.89 / 37.17	7.12 / 31.95 / 50.88	26.64
	PIR	Swin Trans/Bert	10.52 / 26.44 / 39.52	6.83 / 23.97 / 39.09	24.40	19.03 / 41.15 / 53.32	13.72 / 42.48 / 62.88	38.76
	HVSA	CNN/RNN	8.60 / 22.78 / 33.21	5.95 / 22.53 / 34.57	21.27	14.60 / 32.96 / 46.46	10.84 / 38.01 / 57.35	33.37
	KAMCL	ResNet/GRU	8.14 / 22.14 / 33.39	6.35 / 23.70 / 38.87	22.10	14.82 / 29.65 / 43.58	10.53 / 35.8 / 55.84	31.70
	DOVE*	ResNet/GRU	8.66 / 22.35 / 34.95	6.04 / 23.95 / 40.35	22.72	16.81 / 36.80 / 49.93	12.20 / 44.13 / 66.50	37.73
	IEFT	Trans/Trans	4.86 / 19.76 / 34.48	6.77 / 23.88 / 38.15	21.32	14.82 / 33.63 / 49.56	10.98 / 37.61 / 58.25	34.14
	RemoteCLIP	Trans/Trans	16.65 / 35.13 / 47.39	14.73 / **40.14** / **56.96**	35.17	27.21 / 50.22 / 63.93	24.20 / 57.21 / 74.96	49.62
	GLISA*	Trans/Trans	19.52 / **40.44** / **52.28**	14.75 / 39.50 / 55.46	36.99	28.21 / 50.94 / 62.50	20.50 / 54.80 / 72.19	48.41
	FSSN	Trans/Trans	**19.58** / 40.34 / 52.05	**15.73** / 39.89 / 56.77	**37.39**	**30.53** / **52.21** / **65.49**	**25.71** / **58.32** / **75.40**	**51.28**

of the experiments are listed in Tables I and II, and we have highlighted the top results of the classic image–text retrieval networks and RS image–text retrieval networks, respectively, to facilitate a clearer comparison with our proposed FSSN network. According to these tables, the following analysis is provided.

1) Results on RSICD: The comparison results on the RSICD dataset are displayed in the left area of Table I. Among all classic cross-modal retrieval methods, NAAF achieved the highest retrieval results, with an accuracy of 16.20 on the mR evaluation metric. In addition, when we turn our attention to the models in the field of RS, an analysis can be obtained that GLISA exhibits the most superior performance compared to other models. It can also be observed from the table that our FSSN obtains the best retrieval accuracy among all pretrained large models, gaining an improvement of 1.08% to GLISA. In addition, for the text-to-image task, the performance of our FSSN is slightly worse than that of RemoteClip in terms of R@5 and R@10 evaluation metrics. This is because RemoteClip is being trained on a large amount of data, thus acquiring strong generalization capabilities. Nevertheless, FSSN still exhibits higher accuracy than RemoteClip, especially in the R@1 evaluation metric. This is due to the robust capability of the proposed FILF, SIBS, and FSSF modules in processing details and local information when applied to RS data that contains a significant amount of noise.

2) Results on RSITMD: The right area of Table I shows various retrieval results on the RSITMD dataset. As we can observe, the NAAF and RemoteClip output with more competitive precision than other classic and RS retrieval models of 27.81 and 49.62 on the mR evaluation metric, respectively. It is worth noting that our model demonstrates outstanding performance in all tasks, including text-to-image and image-to-text, as evidenced by all evaluation indicators. In terms of the mR evaluation metric, the FSSN improves RemoteClip by 3.35%, and we infer this remarkable result benefits from the reliability and stability of the raised saliency modeling network that considers both the frequency and the spatial domain by introducing FILF and SIBS modules, and realize robust fusion based on the FSSF module.

3) Results on UCM: The results of all retrieval models trained on the UCM dataset are described in the left part of Table II. In this dataset, the MTFN and RemoteCLIP models achieve the best retrieval results in natural and RS models, respectively. In addition, the proposed model in the UCM dataset shows the largest improvement, at 7.58%, compared to RemoteCLIP. This improvement is attributed to our approach of not only using low-pass signals to filter noise in images, but also introducing boundaries to extract robust information in RS data without noise transferring, thereby significantly enhancing the credibility of the retrieval method. However, the R@10 indicator results for the text-to-image task do not exceed those of RemoteCLIP, which is still due to the strong generalization of RemoteCLIP.

4) Results on SYDNEY: We can find the comparison results trained on the SYDNEY dataset in the left part of Table II. The SCAN t2i and RemoteCLIP gain the highest values of 49.05 and 53.33 on the mR evaluation metric in the natural and RS cross-modal retrieval models, respectively. Note that for text-to-image retrieval, the proposed FSSN obtains the most admirable accuracy on all metrics of R@1, R@5, and R@10, while for image-to-text, the model still shows outstanding performance in the R@1 and R@10 metrics, but failed in R@5 compared to IEFT. Although the proposed FSSN cannot occupy an advantageous position on every metric, it still far exceeds all baselines on the mR evaluation metric, improving

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TABLE II
COMPARISONS OF IMAGE–TEXT RETRIEVAL EXPERIMENTS ON UCM AND SYDNEY DATASETS

Type	Method	Backbone Vision/Text	UCM			SYDNEY		
			Image-to-Text R@1 / R@5 / R@10	Text-to-Image R@1 / R@5 / R@10	mR	Image-to-Text R@1 / R@5 / R@10	Text-to-Image R@1 / R@5 / R@10	mR
N	SCAN t2i	ResNet/GRU	13.89 / 45.81 / 68.59	13.11 / 50.69 / 78.21	45.05	19.14 / 50.71 / 74.12	17.81 / 55.94 / 76.57	49.05
	SCAN i2t	ResNet/GRU	12.81 / 46.99 / 69.11	12.59 / 46.89 / 72.71	43.52	20.42 / 54.16 / 67.62	16.13 / 57.61 / 75.98	48.65
	CAMP-triplet	ResNet/GRU	11.20 / 44.29 / 65.72	9.89 / 46.11 / 77.29	42.42	22.83 / 50.46 / 75.88	15.31 / 43.14 / 70.39	46.34
	CAMP-bce	ResNet/GRU	15.09 / 47.18 / 68.59	11.93 / 47.22 / 76.98	44.50	15.64 / 49.65 / 71.33	11.59 / 51.34 / 76.16	45.95
	MTFN	ResNet/GRU	10.68 / 47.62 / 64.31	14.74 / 52.65 / 81.13	45.19	21.71 / 51.58 / 69.03	14.13 / 55.96 / 78.62	48.51
	SGRAF	ResNet/Bi-GRU	4.76 / 36.19 / 63.81	10.76 / 48.67 / 78.38	40.43	15.52 / 43.10 / 62.07	7.93 / 46.55 / 73.10	41.38
	NAAF	ResNet/GRU	12.38 / 49.52 / 64.76	11.52 / 44.38 / 66.29	41.48	24.14 / 56.90 / 68.97	17.24 / 46.90 / 75.86	48.34
R	SAM i2t	CNN/RNN	13.78 / 46.77 / 67.15	11.32 / 49.27 / 81.81	45.02	24.12 / 55.17 / 75.93	13.41 / 54.46 / 76.62	49.95
	SAM t2i	CNN/RNN	11.92 / 45.23 / 70.95	11.42 / 53.41 / 83.12	46.01	27.64 / 56.87 / 72.35	19.01 / 57.93 / 79.96	52.29
	LW-MCR	SqueezeNet/RNN	13.81 / 40.10 / 59.05	10.38 / 44.86 / 74.95	40.53	17.24 / 43.10 / 63.79	16.21 / 56.21 / 75.17	45.29
	AMFMN	ResNet/GRU	14.48 / 51.17 / 67.35	14.49 / 51.82 / 80.71	46.67	20.69 / 50.73 / 75.23	13.45 / 60.00 / 82.70	50.47
	MCRN	CNN/GRU	12.38 / 44.29 / 68.09	11.71 / 50.48 / 92.76	46.62	25.86 / 55.17 / 65.52	15.52 / 54.83 / 80.34	49.54
	CMFM	ResNet/GRU	13.33 / 50.00 / 66.67	11.43 / 49.24 / 74.57	44.21	17.24 / 51.72 / 62.07	15.17 / 55.52 / 81.38	47.18
	HVSA	CNN/RNN	14.29 / 39.52 / 59.52	13.71 / 48.48 / 73.33	41.48	13.79 / 48.28 / 62.07	14.48 / 55.52 / 77.24	45.23
	IEFT	Trans/Trans	11.90 / 47.62 / 75.71	13.63 / 53.03 / 90.45	48.72	22.41 / **63.79** / 72.41	18.59 / 54.65 / 79.55	51.90
	RemoteCLIP	Trans/Trans	15.71 / 50.95 / 80.47	17.71 / 63.33 / **98.00**	54.36	20.69 / 56.89 / 74.14	21.03 / **64.83** / 82.41	53.33
	FSSN	Trans/Trans	**23.81** / **64.29** / **81.90**	**20.76** / **65.71** / 94.38	**58.48**	**24.18** / 60.34 / **75.86**	**23.10** / **64.83** / **85.17**	**55.58**

TABLE III
ABLATION STUDIES

Ablation Model	RSITMD						
	Image-to-Text			Text-to-Image			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
FE	27.21	50.22	63.93	24.20	57.21	74.96	49.62
FE+FILF	26.32	51.32	64.60	24.34	58.32	**76.33**	50.21
FE+SIBS	30.30	50.66	**65.70**	25.88	57.21	74.46	50.70
FE+FILF+SIBS	28.76	**52.43**	64.82	**26.41**	**59.02**	74.07	50.92
FE+FILF+SIBS+FSSF	**30.53**	52.21	65.49	25.71	58.32	75.40	**51.28**

by 4.22% than the current best model RemoteClip. The results on SYDNEY indicate the availability and feasibility of the proposed methods.

D. Ablation Studies

The ablation experiments are performed in this section to further discuss the availability of each proposed module, and the relevant results are displayed in Table III. Note that the ablation studies and all the following experiments are performed on the RSITMD dataset. The symbol FE in the table represents that only the feature extraction is applied in the retrieval network, and FILF, SIBS, and FSSF represent the proposed three modules. "+" means that the model integrates and adopts the corresponding modules. As can be seen from the table, FE outputs the most unsatisfactory accuracy regarding the mR evaluation metric since the training process fails to consider the impact of RS noise on feature representation capability. When the model is integrated with the frequency-domain feature extraction method FILF, the results have been notably improved, surpassing the FE by 1.19%. It is noteworthy that after the introduction of the SIBS module, our model has made miraculous progress, promoting mR by 2.18% compared to FE, benefiting from the effective utilization of boundaries and the establishment of the assessment mechanism to prevent the features in the spatial domain from being interfered by RS noise. As the modules continue to be combined, the accuracy of retrieval is gradually enhanced. When the model applies all three modules, which is shown in the last row of the table, the model shows the best performance with an improvement of 3.35% rather than FE. Therefore, the above data demonstrate the feasibility of our method from a quantitative perspective.

We also explore the semantic localization evaluation index (SeLo) [56] to qualitatively analyze the performance of the proposed modules. SeLo is a visualized evaluation metric designed for cross-modal retrieval tasks. It can output the most relevant RS locations based on query text information. In our experiment, we randomly select three demo sample pairs and input them into various methods, including FE and the fusion of different modules, and generate the salient locations in images according to the query texts, as shown in Fig. 4. The left figure of each demo displays the generated SeLo heat maps, while the right figure presents the superimposed heat maps on the corresponding images. As illustrated in the figure, the proposed modules demonstrate a competitive advantage. For instance, in the last demo, the FE method yields the poorest results, failing to locate even approximate areas based on the query text. However, by integrating the FILF module, the retrieval model can identify the approximate areas in the image. The SIBS also contributes to performance improvement, clarifying some targets in the image. Finally, with all three modules integrated, the model can accurately

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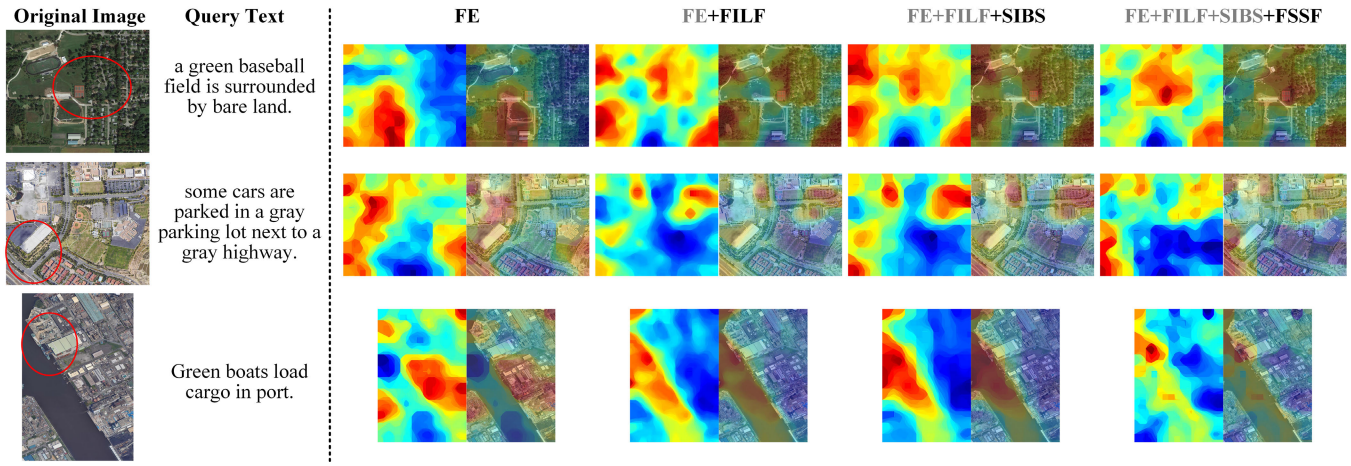


Fig. 4. Visualized qualitative comparison of SeLo on the FE and the fusion of different modules. (Left) Each demo displays the generated SeLo heat maps, while (right) superimposed heat maps on the corresponding images.

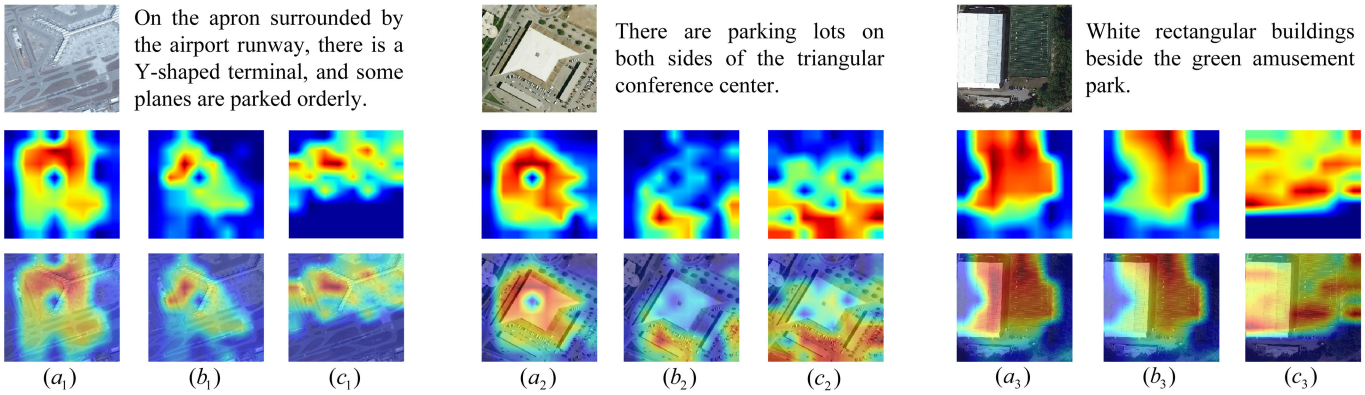


Fig. 5. Visualization of salient masks output by self-attention mechanism, cross-attention mechanism, and SIBS module. (a_1)–(a_3) generated by self-attention mechanism, (b_1)–(b_3) output of cross-attention mechanism, and (c_1)–(c_3) represents the spatial confidence mask m^{vs} generated by the SIBS module. (Top) Image–text matching pairs. (Middle) Salient masks. (Bottom) Superimposed image.

output the most relevant RS locations based on the query text information.

E. Performance of SIBS

As mentioned above, RS data usually contains a lot of noise, which will be transferred in attention-based saliency methods, thereby reducing the representation capability of features. In order to verify the above conclusion, we perform the following three sets of experiments. First, in FSSN, instead of applying the similarity distribution boundaries to explore salient information, we use the self-attention mechanism on image and text features V and U to extract effective features by the self-attention mask. Similarly, the cross-attention mechanism is also employed to obtain salient information through the calculation of the cross-attention mask. Subsequently, the above-generated masks of images are visualized in Fig. 5, where the top row shows the image–text matching pairs, the middle row visualizes the saliency masks of the images, and the bottom row displays the superposition of the saliency mask and the original images. (a_1)–(a_3) are generated by self-attention mechanism, (b_1)–(b_3) are the output of cross-attention mechanism, and (c_1)–(c_3) represents the spatial confidence mask m^{vs} generated by the SIBS module. Through Fig. 5, we can draw the following analysis. By comparing (a_1) and (b_1), we can find that the salient areas of (a_1) are more confused than those of (b_1), thus concluding that cross-attention has achieved superior performance than self-attention. The reason is that cross-attention considers mutual guidance between different models and could make full use of multiple modal clues to mine salient information purposefully. In addition, it should be noted that the visualization results of (c_1) far exceed those of (a_1) and (b_1), such as (c_1) has the capability to focus on the "Y-shaped terminal" more accurately. That is beneficial from the effective utilization of boundaries and the reliable design of the evaluation mechanism, which avoids the interference of RS noise during the saliency extraction process, thereby improving the representation capabilities of features. Similarly, the same conclusion can also be obtained from observing (a_2)–(c_2) and (a_3)–(c_3). Therefore, we believe that the proposed SIBS module has competitive advantages compared with other spatial saliency methods in RS image–text retrieval tasks.

F. Performance of FSSF

In order to prove the effectiveness of the FSSF module, we perform relevant experiments by taking different means

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TABLE IV
PERFORMANCE OF FSSF

Method	RSITMD		
	Image-to-Text R@1 / R@5 / R@10	Text-to-Image R@1 / R@5 / R@10	mR
PA/o	28.76 / 52.43 / 64.82	**26.41** / **59.02** / 74.07	50.92
Con/o	30.08 / **52.87** / **66.59**	23.93 / 56.94 / 74.73	50.86
PA/d	28.09 / 51.32 / 65.26	23.62 / 57.12 / 75.61	50.17
Con/d	30.30 / 51.99 / 65.26	25.92 / 58.49 / 74.73	51.12
Ada/w	30.45 / 52.04 / 64.88	25.84 / 57.66 / **75.67**	51.09
FSSF	**30.53** / 52.21 / 65.49	25.71 / 58.32 / 75.40	**51.28**

to fuse image features generated from the frequency domain and spatial domain. The experimental results are shown in Table IV, where the PA/o refers to point-wise addition without considering the identity mask, and Con/o means the concatenate without considering the identity mask. The PA/d in the table refers to point-wise addition with discarding features with inconsistent masks, and similarly, Con/d denotes concatenate with discarding features with inconsistent masks. Finally, the sign Ada/w represents adaptive fusion with considering identity mask. From the table, we can conduct the following analysis.

The PA/d obtains inferior retrieval performance because discarding inconsistent features will lead to inadequate utilization of effective features. In addition, PA/o and Con/o promote the accuracy of the model, improving by 1.49% and 1.38% than PA/d, respectively. From the above data, it can also be calculated that the concatenate operation is more useful than the point-wise addition. However, failing to address inconsistencies means simply integrating features haphazardly and failing to produce the best results. Thus, we come up with a mechanism to fuse features adaptively, but as shown in the table, Ada/w is not the optimal choice. As we expected, FSSF outperforms other fusion methods, achieving an accuracy of 51.28 on the mR evaluation metric. The reason behind this is that when masks are inconsistent, both concatenation and point-wise addition methods only crudely fuse spatial- and frequency-domain features without considering the inconsistent noise judgment. In contrast, FSSF can utilize the assessment mechanism to automatically select features with high confidence in noise judgment, which acts as a secondary filtering process, ultimately enhancing the robustness of the feature representation.

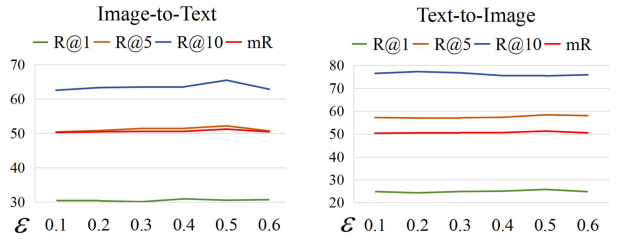


Fig. 6. Performance of hyperparameter ε.

TABLE V
COMPLEXITY ANALYSIS

Type	Method	Parameters	Time(s)
N	SCAN	37M	0.0081
	CAMP	38M	0.0061
	MTFN	69M	0.0057
	SGRAF	28M	0.0085
	NAAF	23M	0.0058
R	LW-MCR	12M	0.0024
	AMFMN	37M	0.0025
	GaLR	47M	0.0043
	MCRN	52M	0.0037
	CMFM	41M	0.0026
	HVSA	34M	0.0025
P	RemoteCLIP	0 M	0.0114
	FSSF	45M	0.0184

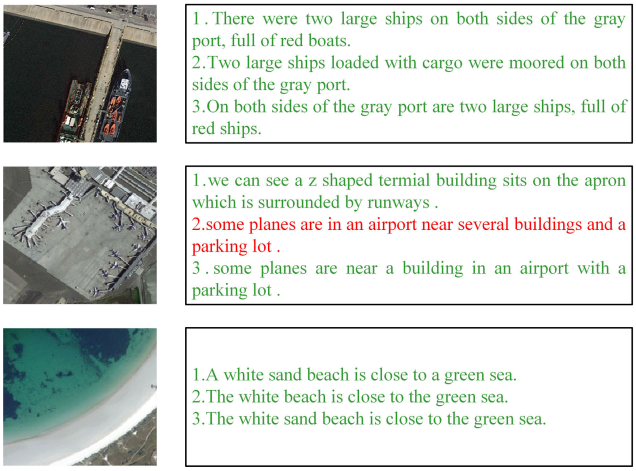


Fig. 7. Top three image-to-text retrieval results on the RSITMD dataset, with green check marks indicating ground truth texts and cross marks indicating wrong results.

G. Performance of Hyperparameters ε

To determine the optimal values of the ε parameter in the FSSF module, we varied ε from 0.1 to 0.6, with intervals of 0.1. The relevant results are shown in Fig. 6 and we draw the following conclusion. For the image-to-text retrieval, it can be seen that as ε grows, the accuracy of the model continues to improve. When ε is set to 0.5, the performance of our model reaches its peak. After that, the precision starts to decrease as ε increases. Note that the red line in the figure represents the mR evaluation metric, representing the highest persuasiveness, which also indicates that the retrieval model with ε taking 0.5 achieves the optimal accuracy. Besides, for image retrieval text, we can obtain similar observations. Therefore, we choose ε to be 0.5 in the proposed network FSSN.

H. Complexity Analysis

We also perform complexity analysis experiments, as demonstrated in Table V. "Parameters" represents the trainable parameters in the network, and "Time" refers to the average time needed to train the network on one sample pair. The table indicates that the large pretrained model, RemoteCLIP, has the fewest trainable parameters compared to models that are not pretrained on a large model. Our proposed model contains 45 million parameters. We attribute

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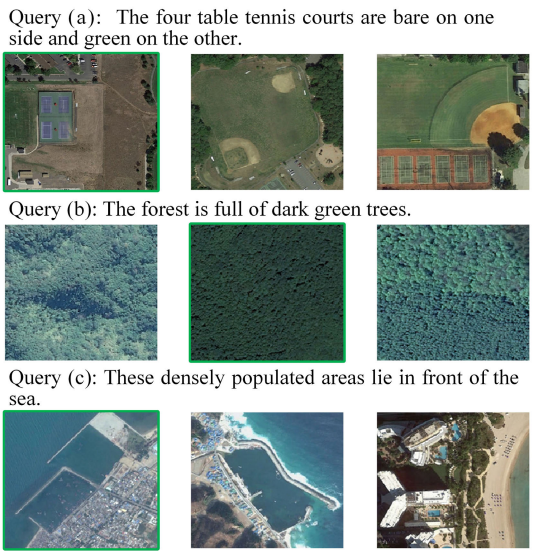


Fig. 8. Top-3 text-to-image retrieval results on the RSITMD dataset, where the green bounding boxes indicate the matched images. (a) Four table tennis courts are bare on one side and green on the other. (b) Forest is full of dark green trees. (c) These densely populated areas lie in front of the sea.

these results to the FILF module, which includes multiple frequency denoising layers, with each layer comprised of several convolution operations. In the future, we plan to reduce the number of frequency denoising layers to decrease computational costs. In addition, our model uses the encoder from RemoteCLIP, leading to slightly longer processing times compared to RemoteCLIP. As a result, our model achieves a balance between efficiency and effectiveness.

I. Qualitative Analysis

In this section, we illustrate some image-to-text and text-to-image retrieval results of the RSITMD dataset in Figs. 7 and 8. It can be seen from Fig. 7 that our model could output the correct retrieval sentence in the majority of cases for the queried image. Despite some mismatched sentences also obtaining high-ranking results, when we pay careful attention to the details of these sentences, we will find that these sentences can also describe the image correctly, which is even difficult to distinguish by humans. For the task of text-to-image, we should note that there is only one correct result in the visualization since each sentence in the RSITMD dataset corresponds to only one image. Although in Query (b), the matched image achieved the second ranking, the image in the first position still conforms to the text description. Therefore, as can be seen from the visualization results, our model shows impressive performance and meets expectations in the cross-modal retrieval task.

V. Conclusion

In this article, we introduce a novel FSSN to handle the cross-modal retrieval task of RS data. A novel idea is raised to realize feature modeling by jointly considering the frequency domain and spatial domain, so as to ensure effective and reliable image–text retrieval in dealing with RS data that contains lots of noise. Based on this, we devise three modules, including FILF, SIBS, and FSSF, where the first two are proposed to extract salient feature representations in the frequency and spatial domain, respectively, while the last is to achieve effective fusion of features obtained from the two domains. Specifically, the FILF fully utilizes low-frequency signals to filter noise in images by exploiting multiple frequency denoising layers, while the SIBS applies the similarity distribution boundaries of matched and mismatched sample pairs, and devises novel image and TAMs to distinguish the effective salient information in images and texts without noise information transfer. Finally, FSSF is designed to effectively integrate the spatial- and frequency-domain features according to the consistency of the identity mask.

Numerous experiments are implemented to verify the performance of FSSN on four large-scale RS cross-modal retrieval datasets and the experimental results reveal that our model outperforms the other 12 baselines. Specifically, the proposed FSSN outperforms the best model RemoteClip by 7.58% of the mR evaluation matrix on the UCM dataset. The FSSN model is a high-quality cross-modal retrieval system that can be effectively utilized in various tasks involving different modalities. It excels in providing robust feature representation and precise alignment between images and text. For example, in land use classification, RS images can be matched with terms like “commercial area” and “green space” for efficient classification. In disaster response scenarios, text queries such as “mudslide” and “flood” can help quickly locate affected areas in satellite images. However, the proposed method still has some limitations. One key issue is the number of parameters, which impacts its efficiency. Our analysis indicates that this is due to the FILF module, which uses multiple frequency denoising layers, and each layer consists of several convolution operations. In the future, we plan to reduce the number of frequency denoising layers to decrease computational costs.

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